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Multi-classification of breast cancer histopathological image using enhanced shallow convolutional neural network

Musa Yusuf^{1,2*} , Armand Florentin Donfack Kana¹, Mustapha Aminu Bagiwa¹ and Mohammed Abdullahi¹

*Correspondence:
yusufmusa@binghamuni.edu.ng

¹ Department of Computer Science, Ahmadu Bello University, Zaria, Kaduna, Nigeria

² Department of Computer Science, Bingham University, Karu, Nasarawa, Nigeria

Abstract

Cancer of the breast popularly known as breast cancer (BC) is the second and third utmost cause of mortality among women in Nigeria and globally, respectively. Biopsy histopathological images (BHI) have gained more attention for the early clinical diagnosis of BC. However, the clinical examination and diagnosis of BC histology images are subject to human error. Consequently, several computer-aided diagnoses (CAD) solutions have been presented to aid histopathologists with the automated classification of cancerous tumor cells on histological images. Deep convolutional neural networks (DCNN) have been utilized to build a sizable portion of the cutting-edge proposed solutions. However, due to the architectural structure of DCNN, which extracts features automatically along with training processes and is coupled with overlapping nucleic features on breast histology images (BHI), the existing solutions suffer from high computational utilization, extensive training time leading to longer convergence times, and reliance on available high-end system resources to build adequate BC classification solutions. In this paper, an enhanced shallow convolutional neural network (ES-CNN) has been proposed for multi-classification of BHI, aimed to improve classification performance and reduce training time across eight BC types and four magnifications in the BreakHis dataset. The research objectives were achieved in three ways. First, we designed the proposed network's architecture, guided by magnification and patient dependencies. Secondly, we implemented a multi-classification model based on the proposed network, and, finally, two categories of experiments were conducted based on classification accuracy and computational utilization. The experimental results revealed that the proposed methods have minimal computational utilization and improved classification performance compared to the existing work. This research reports a multi-classification accuracy of 96%, 95%, 98%, and 96% across 400×, 200×, 100×, and 40× image magnifications, respectively.

Keywords: Breast cancer, Deep learning, Shallow convolutional neural network, Computer vision, Medical image analysis, Machine learning, Lightweight model

Introduction

Breast cancer (BC) is the most prevalent malignancy in women and a significant worldwide public health issue [1]. As shown in Fig. 1, the World Health Organization (WHO) reported BC as the deadliest cancer disease in Nigeria and is responsible for 11,564

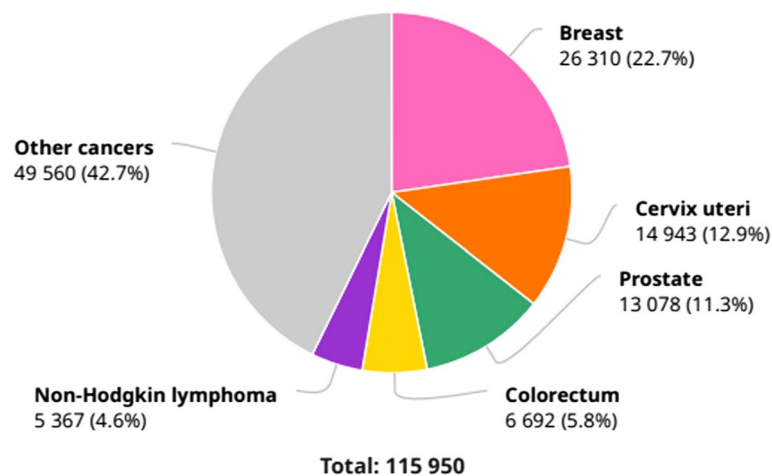


Fig. 1 Nigerian cancer cases 2020: male and female cases, across all ages [4]

deaths, making 16% of total deaths per annum, with an incidence rate of 22.7%. According to Siegel et al. [2], the African BC average mortality rate is also on the increase. A report by Amin et al. [3] estimated the mortality rate of BC in Nigeria as 33.2% (that is 459,000 in 1,384,155 cancer cases) annually, making Nigeria one of the top countries in Africa with the highest number of BC cases recorded annually.

BC clinical screening and monitoring methods include positron emission tomography (PET), mammography [5], breast ultrasound [6] biopsy or histology, magnetic resonance imaging (MRI), thermography, surgical incision, and ultrasound. Histology involves a pathologist microscopically examining a stained whole slide image (WSI) to study nuclei features that characterize the disease-related areas on the biopsy tissue section being examined. Because of the comparable clinical presentations of different forms of cancer, interpreting the data generated from these procedures is extremely challenging. Data analysis is laborious and time-intensive, but it is also a crucial step in making a differential diagnosis [7]. The examination result serves as a basis for oncological decision-making and further clinical treatment, which is delicate for patients' survival [8]. The serial examination for BC metastasis is subject to subjectivity and the expertise of the pathologist due to its tediousness and time-consuming [9, 10]. With the mentioned challenges clinicians face, computer-aided diagnostic (CAD) solutions have surfaced in recent years to tackle the challenges of manual examination of breast histological images. CAD over the years proved to have greater benefits compared to the traditional histopathological microscopic examination.

Among the proposed CAD solutions, deep learning (DL) is generating a lot of interest because of their autonomous feature extraction and representation learning abilities in the classification of BC [11–17]. Another reason for its appeal is the advancement in processing power [18]. CNN, a common DL network variation, is well recognized for its weight-sharing and local connectivity properties and has several applications in computer vision [19]. However, they use a lot of important computer and memory resources, as well as a lot of training time [20]. This paper, therefore, proposed an enhanced shallow convolutional neural network (ES-CNN) to further reduce training time with improved classification performance.

Shallow CNN

DCNN models are well-known for the glitch of overwhelming time-consuming training, although they perform well in computer vision (CV)-related applications, and the research effort on SCNN has been explored recently [10, 21–23]. In the work of Islam et al. [22], SCNN was proposed for learning neuromuscular activity recognition using instantaneous high-density surface electromyography (HD-sEMG) images from scratch. Their model aimed to reduce design complexities in the state of the art with competitive performance. Their model expects an input of 16×16 grayscale images followed by three 3×3 convolutional layers (CL) with 64 filters each. Next is 3×3 CLs with 128 filters each. A fully connected layer of 128 neurons was introduced before taking the global average over 8×8 spatial dimensions followed by the softmax. However, their network seems to be deeper than the proposed frame in this paper. They performed comparative experimentation on five different activation functions and reported expected linear unit (ELU) as the best choice for their proposed model against ReLu, leaky-ReLU, Sigmoid, and max-pooling.

The work of Gorokhovatskyi and Peredrii [21] used the CIFAR and MNIST datasets among other 4 datasets that were tested on 8 CNN layers comprised of $2 \times 3 \times 3$ CLs with 32 filters each, $2 \times 2 \times 2$ max-pooling layer (MaxPL), 2 dropout layers, and a fully connected layer 512 neurons. It took their model 3 h and 20 min to train the SAT-4 dataset which is an airborne dataset of four classes of sizes $28 \times 28 \times 3$. Even though their training was done on a 3.6-GHz processor running on GPU, with 16 GB, their model took over 3 h to train on the mentioned datasets.

Zhang et al. [24] performed implicit discourse relation recognition using SCNN which contains only one hidden layer trained in one PDTB 2.0 corpus; they tested their model against a support vector machine, transductive SVM, and recursive auto-encoder in which their model outperformed the three baseline models. Lei et al. [20] propose a novel SCNN model to tackle the blind image sharpness assessment (BISA) problem using one feature extraction layer that employs local contrast normalization by local variable normalization and displacement of localized mean by introducing orientations that monitor non-linearity. They retrieved features with support vector regression and achieved the best overall performance, indicating a high correlation with subjective judgment. To improve accuracy in computer vision tasks, Liu et al. [25] offer SCNN that uses global averaging pooling that performs convolutional operations with filters that logarithmically scan over input images. They put their model through its paces and performed two evaluations: item classification using the CIFAR-10 dataset and capability for facial expression recognition in which the multi-PIE dataset was used.

According to Li et al. [14], dealing with SCNN may require the identification of tampered parts in images. Their proposed model was employed as the fundamental detection component to detect and locate the tampered element in an image. On the CASIA 2.0 datasets, their model works well with all picture types. With 2 CLs and a dropout layer before the flattening layer with 25,088 neurons, their model takes in 32×32 patches of CbCr color type as input and predicts using the sigmoid activation function.

Montaha et al. [10] proposed a resilient and effective SCNN architecture named MNet-10. They attain over 97% classification performance across eight medical datasets with different modalities with a minimal level of computational complexity. The

incorporation of augmentation methods such as elastic deformation contributed to the performance of their model. They further use transfer learning techniques to compare the performance consistency of the MNet-10. The results of their study may provide researchers with an analysis of medical data for ablation investigations and augmentation procedures. It can be inferred from their study that SCNN and data augmentation methods can work together to help deal with a small number of images, and training time and complexity can be reduced with the aid of shallow design. However, the data augmentation method used may increase the training time of MNet-10.

Deep CNN

One of the methods for dividing the large histology image into small batches and then using it to train CNN was proposed by Spanhol et al. [26]. The classification result is generated by combining the patch result for the whole slide image. Spanhol et al. [27] presented a BC histological classification approach, which makes use of deep features extracted from a CNN. Features of previously trained CNN are used as feature extractors. They extracted deep feature vectors using DeCAF features and then used them as an input to a traditional classifier as any other features. DeCAF is a transfer learning technique used in the classification process to handle underperformance due to short training and twining complexity. They evaluated their model performance based on image and patient level. In the patient-level accuracy, they achieved an accuracy of 86.0%, the best in $200 \times$ magnification, while in the image-level accuracy, they achieved an accuracy of 84.7% as the best result. Based on the recommendation of Spanhol et al. [26] instead of training models with problems with hundreds of images, weights trained on millions of images can be used for training and classification of the small dataset problem. This method is referred to as transferred learning (TL). Hence, DCNN offers an accurate and faster means of identifying invasive cancerous diseases in medical image analysis classification compared to ML.

A three-layered CNN model for WSI histology analysis proposed by Spanhol et al. [27] compares their work with well-handcrafted features. They evaluated the reliability, confidence, speed, and reproducibility of their classifier using several strategies for training the CNN architecture based on extracting patches obtained randomly or by sliding window mechanism. They reported an accuracy of 84.23% and an F-score of 71.80%. As a result, there was an improvement of 5% on the $40 \times$ magnification compared to their previous methods. A hybridized multi-classification approach by Araújo et al. [28] employed CNN to classify BHI into four classes. The feature set extracted from CNN was classified with support vector machine (SVM). They did not consider searching for an aggressive area of interest.

Bayramoglu et al. [29] presented two CNN-based magnification-independent architectures to classify BHI into malignancy and benign classes. The first architecture is named multi-task CNN used to predict both the malignancy and magnification feature, while the second is single-task CNN used for predicting malignancy. Their method is complex in terms of reproducibility. Although they achieved 83.7% classification accuracy on the $200 \times$ magnification, 83.02% was achieved from the evaluation of the $400 \times$ image [30] address the issue of subordinate classes in BC using a structured deep-learning model. Due to the structural nature of their methods, they lacked

flexibility when extending them. However, in the implementation, they achieved better accuracy compared to the method presented by [28, 29].

A novel DNN technique guided by structural and statistical information derived from BHI was proposed by Nahid et al. [31]. In the preprocessing phase, their methods employed K-means clustering and mean-shift clustering algorithms before classification. They proposed three models for classification: the first model used CNN for classification at the classification phase, the long short-term-memory (LSTM) network, and the hybridized model employs CNN + LSTM for breast histopathological image classification was implemented in the second model. SVM and softmax were compared as classifiers in their proposed solution.

Salunkhe and Patil [32] introduce a comprehensive multistage pipeline combining advanced preprocessing (weighted GIF and TOGC) and classification with Rapid Tri-Net. While the method achieves a robust approach by integrating texture and color features with a dual-stage segmentation, it suffers from computational overhead. The SMC-based segmentation and the optimization-heavy feature selection via SDDO add significant complexity, potentially limiting scalability to real-time clinical workflows. Furthermore, while the method focuses on feature richness, the balance between computational cost and incremental accuracy gains is inadequately addressed.

The low-weight parameter systems presented by [33] effectively minimize computational requirements while maintaining high accuracy (up to 98.64% with ViT). These lightweight networks, leveraging hybrid ViT-CNN architectures, demonstrate efficiency in training time and parameter optimization. However, splitting images into patches for localized feature extraction risks losing contextual information, especially when applied to datasets with heterogeneous tumor morphologies. This trade-off may undermine the generalizability of the models to other datasets or clinical settings, which are critical for broad application.

Balasubramanian et al. [34] propose an ensemble approach incorporating VGG16 and ResNet50, achieving impressive accuracies for both BACH (95.31%) and BreakHis (98.43%) datasets. While the ensemble strategy and novel image patching techniques improve precision, however, multiple high-capacity models are reliant on increased computational complexity. Such architectures may require substantial computational resources for training and deployment, rendering the approach less feasible for resource-constrained environments like low-budget labs.

The deep belief network (DBN) proposed by Karthiga et al. [35] addresses dormant factor issues using diversity-promoting priors. Their work contributes to improved classifier performance to 98% accuracy. However, DBNs inherently suffer from high training costs and sensitivity to hyperparameter tuning, as noted in the study's reliance on extensive epochs and careful parameter adjustments. These constraints suggest limited scalability and potential overfitting risks, especially when applied to new datasets beyond BreakHis and Kaggle.

Pal et al. [36] achieve notable accuracy with vision transformer (ViT) models, reporting remarkable performance in binary classification. However, their eight-class classification results reveal a decline in metrics, highlighting the challenge of scaling ViT-based methods for multi-class scenarios. While the study introduces ensemble models known as ViT-ConvMixer, the reduced performance of these ensembles

underscores inefficiencies in integrating multiple architectures, potentially due to overfitting feature extraction.

Ramasamy et al. [37] conducted a performance analysis of transfer learning models, identifying DenseNet121 with an accuracy of 96.5% among several other methods. While transfer learning effectively utilizes pre-trained knowledge, it relies heavily on fine-tuning which may hinder the applicability of their model across datasets of different magnifications. The computational demands of fine-tuning models like DenseNet121 could be excessive for rapid model deployment.

Despite the strengths of the reviewed literature, most of the studies present a few frequent limitations. A common challenge has to do with ascertaining computational complexity and classification accuracy. Techniques such as multistage pipelines, ensemble strategies, or hybrid models, despite improving precision, often introduce computational burdens that consequently limit scalability and real-time applicability in clinical settings. Another critical gap is the limited exploration of lightweight models that can efficiently process high-dimensional histopathology data without compromising accuracy, which ES-CNN aims to close this particular gap. Although methods like ViT-CNN hybrids [33] and regularized DBNs [35] show reasonable performance in this direction, further contributions are needed to optimize computational resource utilization.

Several studies considered BreakHis [38] as benchmark datasets that presented patient- and magnification-based histopathological image datasets for binary and multi-classification BC [39]. So far, to our knowledge, this work is the first work that has conducted and reported multi-classification based on eight (8) BC types. Most of the existing studies present magnification-based results. The major contributions of this paper are presented as follows:

1. Design of an architecture of the proposed ES-CNN for the multi-classification of BHI that requires fewer weights and less computation time for training breast histopathological images.
2. The proposed architecture is used to implement a multi-classification model concerning magnification and patient dependence.
3. An experiment was performed to explore the relationships between the classification performance and computational time of the ES-CNN model based on patient and magnification dependence.
4. The evaluation result of the ES-CNN model on the BreakHis shows that ES-CNN achieves better classification performance with lower computational utilization compared to the existing work.

Methods

This study aimed to build an ES-CNN model for efficient multi-classification of cancerous tumors on breast histopathological images with reduced computational efficiency and improved detection performance. The performance of the proposed was performed in two phases: patient-based and magnification-based evaluation.

This model uses the publicly available BreakHis image corpus made public by Spanhol et al. [38] as the benchmark dataset for developing and evaluating the proposed ES-CNN model. The BreakHis dataset was prepared and released in

collaboration with Pathological Anatomy and Cytopathology and P&D Laboratory, Paraná, Brazil. The benchmark dataset comprises a total of 7909 microscopic images of breast biopsy, majorly classified into 2 categories. The benign categories consist of 2480 images, while the malignant categories consist of 5429 breast tumor images as shown in Table 1. The histological types are categorized into four malignant tumors (breast cancer): carcinoma (DC), lobular carcinoma (LC), mucinous carcinoma (MC), and papillary carcinoma (PC) and four benign, namely adenosis (A), fibroadenoma (F), phyllodes tumor (PT), and tubular adenoma (TA).

The magnification-based samples of $400\times$, $200\times$, $100\times$, and $40\times$ of the BreakHis dataset are presented in Fig. 2, while the samples of patient-based categorization either benign or malignant are presented in Fig. 3.

Table 1 Features of BreakHis dataset showing details of the patient examined [38]

Magnification	Benign					Malignant					Total per magnification
	A	F	PT	TA	Total	DC	LC	MC	PC	Total	
40 ×	114	253	109	149	625	864	156	205	145	1370	1995
100 ×	113	260	121	150	644	903	170	222	142	1437	2081
200 ×	111	264	108	140	623	896	163	196	135	1390	2013
400 ×	106	237	115	130	588	788	137	169	138	1232	1820
Total	444	1014	453	569	2480*	3451	626	792	560	5429*	7909*
No. of patient	4	10	3	7	24	38	5	9	6	58	82

* Indicated the total number of benign images, the total number of malignant images and the of total number of both malignant and benign images in the BreakHis datasets used in the evaluation of ES-CNN

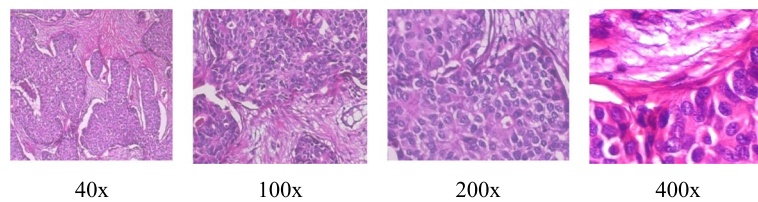


Fig. 2 BreakHis image magnification-based samples

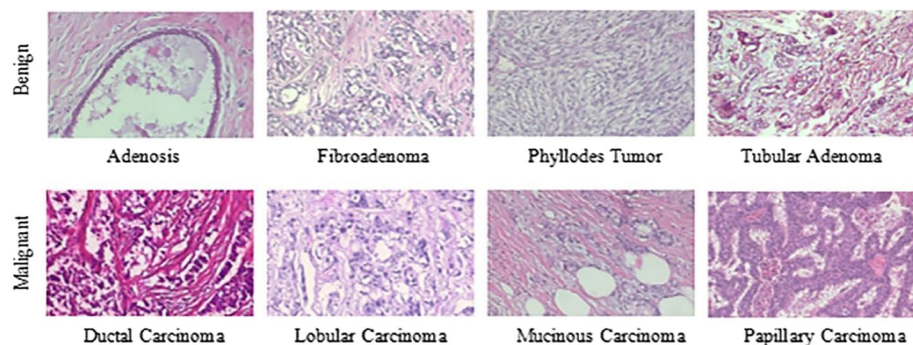


Fig. 3 BreakHis patient-based samples representation as benign and malignant tumor categories

Evaluation metrics

The confusion matrix (CM) is a tabulation used for classification model performance evaluation and for generating evaluation metrics. In this paper, Eqs. 1, 2, 3, and 4 are the evaluation metrics that were generated from the CM.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (1)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{FN} + \text{FP} + \text{TN} + \text{TP}} \quad (3)$$

$$\text{F1 - Score} = \frac{2 * \text{recall} * \text{precision}}{\text{precision} + \text{recall}} \quad (4)$$

ES-CNN framework

In Fig. 4, the proposed architecture for the ES-CNN model is presented. It starts with the source dataset used in this research, followed by data preprocessing also known as feature extractions (FE), the classification layer, and finally probabilistic distributions layer. Details of what each layer represents are discussed as follows.

1. The input layer (IL): $299 \times 299 \times 3$ is the expected input image size meaning $(299 \times 299 \times 3)$.
2. The CL: This layer is responsible for the effective extraction of the nucleic and overlapping feature patterns on the histopathological images. Features were extracted using different convolutional kernels. The ES-CNN consists of filters of sizes 16, 32, and 64 used in the 2×2 CLs.
3. MaxPL: The computational complexity and dimensions were reduced using three 2×2 MaxPLs while maintaining valuable features that were extracted from the CLs nearly intact. The MaxPLs output a matrix that has an equal size of input and output channels. The pooling is responsible not only for improving the training speed and

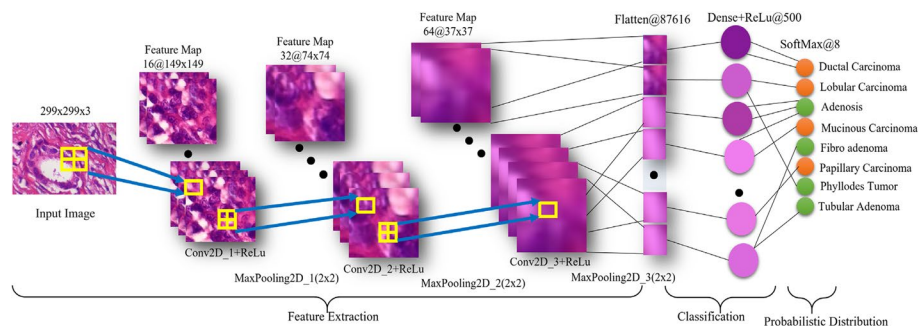


Fig. 4 Framework for ES-CNN

reducing the risk of overfitting but also for reducing the redundancy of data features of the input images.

4. The fully connected layer (FCL): The layer is essentially $1 \times 1 \times 500$ convolutional operations that were done with the same convolution kernel size and the output from the previous layers. In this layer, all the nodes are connected to the node before it.
5. Probabilistic distribution layer: This layer is responsible for the multi-classification of the features from the 1×1 layer of the FCL. The multi-classification is achieved using softmax by converting each output generated from the model into probability distribution and making conclusions based on the obtained probability.
6. The proposed ESCNN architecture design addresses the challenges faced by DCNNs by reducing the hidden layers as well as introducing early stoppage as a regularization technique whenever the validation loss begins to depredate to improve computational efficiency. These have reduced computational convolutions implemented at the feature engineering layers during the training phase as shown in Fig. 4 while reducing the training time with sturdy performance. Unlike DCNNs, which rely on multiple deep layers requiring extensive computation and memory, ES-CNN makes use of fewer layers but carefully optimized convolutional filters for feature extraction without overburdening resources by relying on the regularization technique used. This shallow design reduces the complexity of the proposed network, enabling faster training thereby putting the proposed ES-CNN as a better inference model, particularly on devices with limited computational power. Based on the stated techniques, ES-CNN was able to achieve significant performance gains without the depth-intensive approach of traditional DCNNs.

ES-CNN algorithm

Algorithm 1 illustrates the steps involved in the classification of BHI with the proposed model. Line 1 is the outline expected size of the proposed model. Lines 2 and 3 handle the magnification and patient categories used for magnification-based and patient-based experimentations respectively. Line 4 to line 17 are responsible for feature extraction and classification. Line 4 checks if there are still images in the input directory. Line 5 is used to loop through all images in the dataset, while line 6 is used for splitting the input dataset into a 20% test set and an 80% train set. Mapping of both the input and test images is done using line 7 to make the data set trainable. The loop on line 8 is used for looping through every image in the train and test set. Line 9 is responsible for forwarding and backpropagation steps (with Eq. 2.5) for feature extraction using the convolutional and MaxPLs.

Training commences from line 10, where the training epochs are declared. On line 11, for each epoch, the model learns for a specified batch size which is a hyperparameter set before the training commences. While training the model computes the accuracy of the training and testing is calculated using line 12. On line 13, the training and testing loss are computed using the cross-entropy loss described in Eq. 2.30.

Algorithm 1 ES-CNN algorithm

```

1. input: BreakHis dataset (299,299,3), Number of image = N
2. magnification_image = dataset_(40x, 100x, 200, 400x)
3. patient_image = dataset_(A, F, PT, TA, DC, LC, MC, PC)
4. while  $N > 0$ 
5.   For  $IMG \in input$ 
6.     batch_train_x, batch_test_labels = trainsplit (input, 80, 20, randomize)
7.     train_data = (X: batch_image_x, Y: batch_image_labels)
8.     For image  $\in$  (train_data, test_data);
9.       feature_extract = (forward & back_prop with eqn 2.5)  $\forall$  (CL & MaxPL)
10.      For  $e \in epochs$ 
11.        For batch  $\in$  batch_size calculate the following for each epoch e
12.           $training\_acc, test\_acc = \frac{TP+TN}{TP+FP+TN+FN}$ 
13.           $training\_loss, test\_loss = -\frac{1}{n} \sum_{i=0}^N \sum_{c=1}^C 1_{y_i \in C_c} \log P_{model}[y_i \in C_c]$ 
14.           $optimizer = w_{s-1}^i - \frac{\eta}{\sqrt{v_s + \epsilon}} \cdot \hat{m}_s$ 
15.          train_history = compute (training_loss, training_acc, test_loss, test_acc)
16.          if new_train_acc is attain
17.            Update and Save weight
18.          Else
19.            Continue with training
20. Return weight

```

The Adam optimizer responsible for controlling the learning rate is presented on line 14. The variables responsible for keeping track, updating, and storing the train and test accuracy as well as train and test loss are in line 15. The mechanism introduced in line 16 and line 19 reduces the computational time taken as a result of updating and saving every accuracy and loss of all train and test images which is the case in the existing study. This mechanism is responsible for making the model only make and update when a better accuracy is attained during the training process else no update and save of the accuracy and loss weights. Finally, the weight of the best patterns of the features extracted from the dataset is returned using line 20 (Figs. 5 and 6).

Results and discussion

In this section, the experimental results and their analysis are presented. The implementation and experimentation of ES-CNN were achieved on a laptop computer with the following: 11th generation central processing unit, Intel Core i7 at 2.50 GHz, Intel(R) Iris(R) Xe graphics, 16-GB primary memory, and 512 SSD. Libraries compatible with Python 3.8.8 were used in the implementation of the proposed model. A total of 20% of the BreakHis dataset was reserved for testing the model, while 80% was used to train the proposed model. The input image resolution of the proposed model is $299 \times 299 \times 3$ which gave the best performance [25]. After setting a mini-batch of size 32 and running the training model for 15 epochs, 43,823,052 trainable parameters were generated. The model performance was evaluated based on four metrics: classification accuracy, F1 score, recall, and precision. However, only the result of the F1 score was reported since it is computed with the outcome of recall and precision. These values were calculated using CM and presented in Fig. 7. The evaluation results of the proposed model on the BreakHis dataset for each of the eight BC types are presented in Table 2.

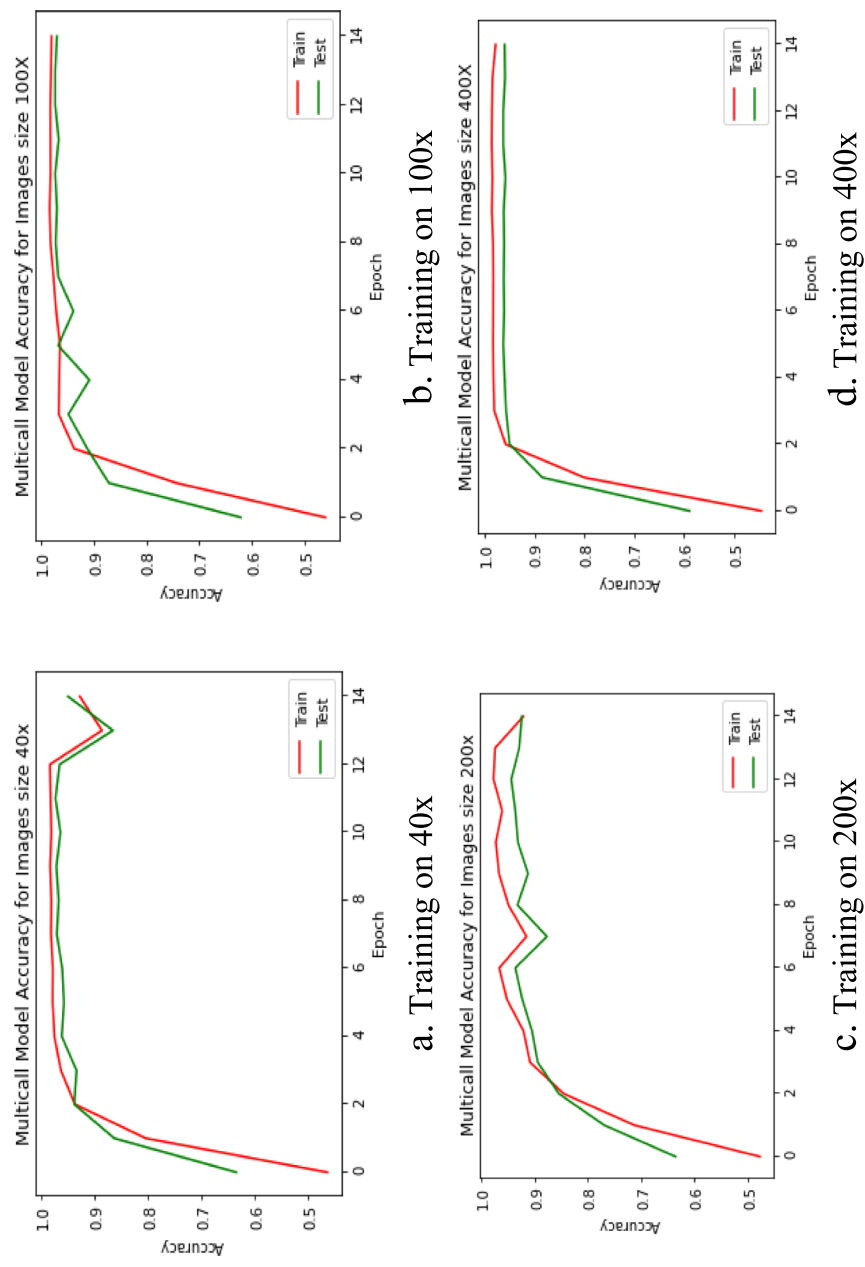


Fig. 5 Magnification-based training accuracy curve for the ES-CNN

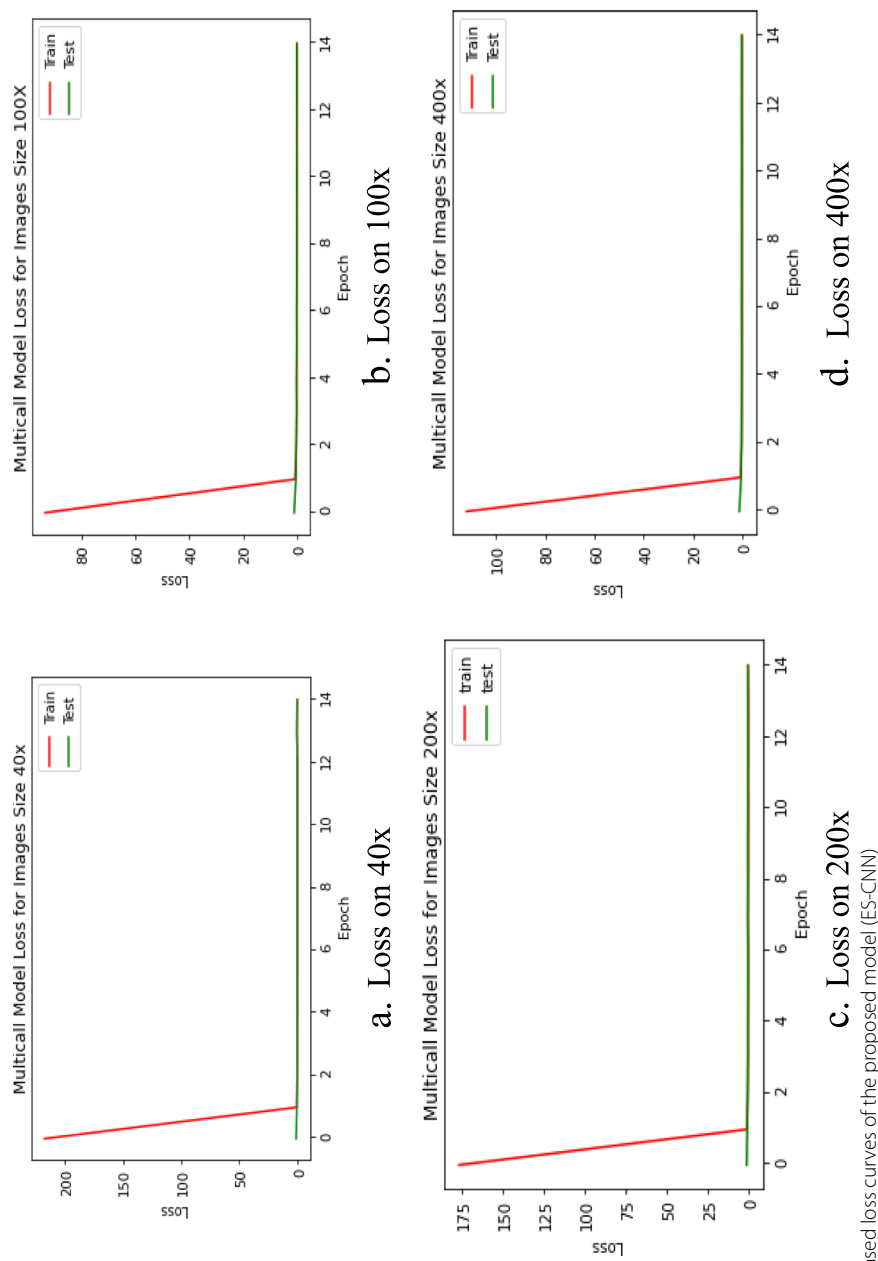


Fig. 6 Magnification-based loss curves of the proposed model (ES-CNN)

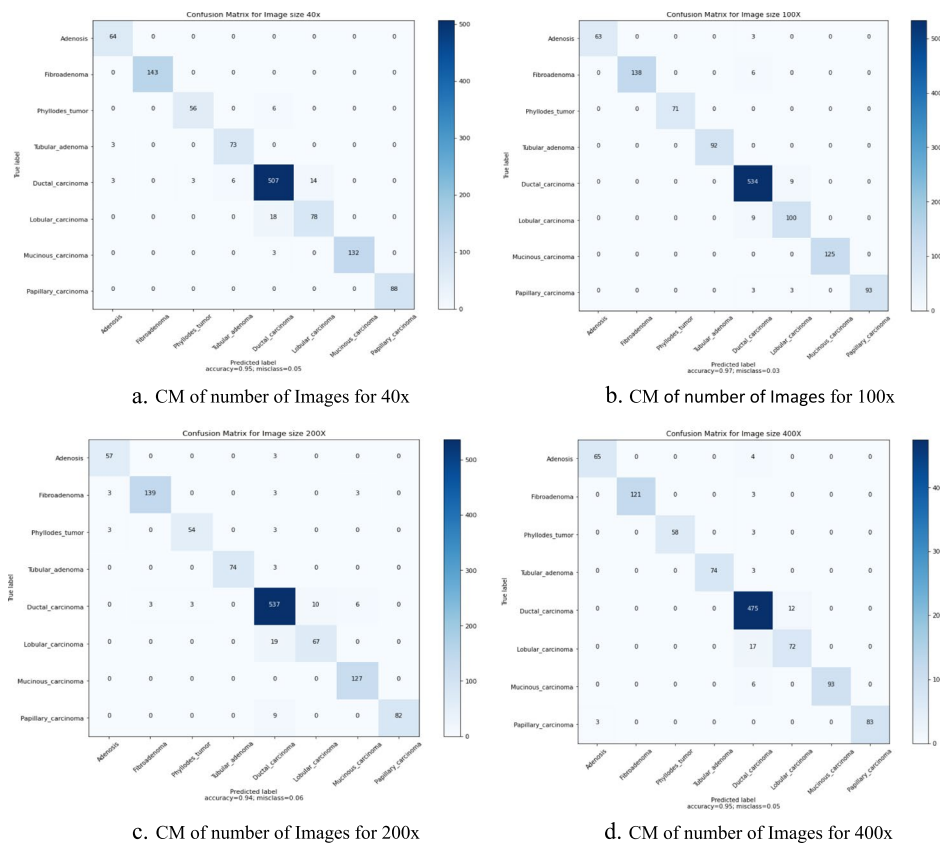


Fig. 7 Confusion matrix result for the images multi-classification of the BreakHis dataset

Analysis of patient-based and magnification-based multi-classification performance

Table 2 presents the calculated score of the F1 score, recall, precision, and multi-classification accuracy. The multi-classification accuracy for magnification-based classification for $400\times$, $200\times$, $100\times$, and $40\times$ is 96%, 95%, 98%, and 96%, respectively. The best precision score is recorded on $100\times$ with 98% followed by $40\times$ with 96%; however, $200\times$ and $400\times$ have the same precision of 95% each. The best recall score is recorded on $100\times$ with 97%, followed by $40\times$ and $400\times$; both have the same recall 96%, while the magnification with the least recall is $200X\times$ with 92%. The magnification with the best F1 score is $100\times$ with 98%, next is $40\times$, followed by $400\times$, and $200\times$ has the lowest F1 score with 94%.

Analysis of training accuracy and training loss

The curves for accuracies and losses accumulated during the training and testing processes of evaluating the ES-CNN model with BreakHis dataset for 15 epochs are presented for each of the four magnifications considered in this paper are presented in Figs. 5 and 6, respectively. The training curves for $400\times$, $200\times$, $100\times$, and $40\times$ are presented in Fig. 5d, c, b, and a, respectively. Figure 5a shows the training and testing curve of the ES-CNN model on images with $40\times$ resolution. It was observed that the proposed model started overfitting when on the 12th epoch; however, there was a surge afterward which could be a result magnification size of $40\times$ BHI. This implies that the proposed

model would need fewer epochs to train the image of $40\times$ compared to the other three magnifications (Table 3).

Figure 6 presents the training and testing accuracy curves for $400\times$, $200\times$, $100\times$, and $40\times$ presented in Fig. 5d, c, b, and a, respectively. Each of the figures shows a uniform rise as the number of epochs increases. Moreover, the decrease in loss curves in training and testing curves for $40\times$, $100\times$, $200\times$, and $400\times$ is also presented in Fig. 6a, b, c, and d, respectively. The result in Figs. 5 and 6 implies that the proposed ES-CNN has a good learning rate for both the training and testing process. In addition, it has the right decreasing order for both training and testing losses as expected.

Analysis of computational time

The report of the processing time of the proposed model experimented on the system whose specifications are presented in the “Conclusions” section of this paper. The ES-CNN spent the highest training time on the $100\times$ magnification, while the least time was on training the $40\times$ magnification. The reason why images of $100\times$ magnification took significantly higher computational time compared to other magnifications was due to the number of $100\times$ magnifications as shown in Table 4. It implies that the higher the image, the higher the computational time of the proposed model. However, the classification accuracy of ES-CNN is higher on the $100\times$ magnification. Based on this analogy, it can be deduced that the performance accuracy of ES-CNN improves with an increase in the number of training sets even though the computational time increases. It is worth mentioning that the ES-CNN model can trade computational time for classification accuracy. Hence, the suitability of the proposed ES-CNN on lightweight devices. The model’s average processing time on Google Collaboratory was averagely less than 4 min even though not reported in this paper.

The BreakHis dataset used for evaluating the performance of the project ES-CNN has several limitations such as the total number of elements in each magnification as well as the patient-based categorization. The size may affect the generalizability of the result of the proposed model. Moreover, changes in imaging conditions, like lighting or staining techniques, could cause biases that might make the network perform worse in generalizing across real-world settings.

Justification of model performance

The percentage representation of the confusion matrix prediction performance of the ES-CNN model is presented in Fig. 7. This result shows how the ES-CNN model has performed in terms of patient-based multi-classification accuracy. The result also illustrates malignancy and benign classification of the eight (8) patient categories in which benign malignant comprises four BC types each. Figure 7a shows that the proposed model is 100% accurate in predicting adenosis (64), ductal carcinoma (507), fibroadenoma (143), and papillary carcinoma (88) of the $40\times$ image magnification of the BreakHis. The lowest patient-based multi-classification reported by the ES-CNN model on the $40\times$ magnification is 78 (78%) which is the lobular carcinoma BC type in which images in this category were misclassified. This means that the proposed model has misclassified 18 (23%) of the patches of the lobular carcinoma BC type which is the lowest prediction

Table 2 Patient-based and magnification-based multi-classification performance of ES-CNN on BreakHis dataset

Patient	40 × (%)			100 × (%)			200 × (%)			400 × (%)		
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec	F1
A	1.00	1.00	1.00	1.00	0.95	0.98	0.90	0.95	0.93	0.92	0.96	0.94
F	0.96	1.00	0.98	1.00	0.96	0.98	0.98	0.94	0.96	1.00	1.00	1.00
FT	1.00	0.90	0.95	1.00	1.00	1.00	0.95	0.90	0.92	0.95	0.95	0.95
TA	1.00	0.96	0.98	1.00	1.00	1.00	1.00	0.96	0.98	1.00	0.96	0.98
DC	0.97	0.95	0.96	0.96	0.98	0.97	0.93	0.96	0.95	0.97	0.97	0.97
LC	0.85	0.90	0.87	0.89	0.92	0.90	0.87	0.78	0.82	0.81	0.89	0.84
MC	0.98	0.98	0.98	1.00	1.00	1.00	0.93	1.00	0.97	0.97	0.94	0.95
PC	0.94	1.00	0.97	1.00	0.94	0.97	1.00	0.90	0.95	1.00	0.97	0.98
Ave. (%)	0.96	0.96	0.96	0.98	0.97	0.98	0.95	0.92	0.94	0.95	0.96	0.95
Acc%	0.96			0.98			0.95			0.96		

in all the 4 magnifications. Furthermore, only 3 out of 136 mucinous carcinoma were misclassified as ductal carcinoma. The ES-CNN model struggled in the classification of lobular carcinoma across all the magnifications, although a prediction of 100 (92%) was reported on the 100 × image magnification as shown in Fig. 7b. The difficulties faced by the ES-CNN model in the prediction of lobular carcinoma are a result of the overlapping characterization of the whitish region found on the lobular carcinoma patches which are visually and obviously different compared to the other seven BC patches shown in Fig. 3.

The best patient-based multi-classification performance established by the ES-CNN model among the four image magnifications is the 100 × magnification. The CM result in Fig. 7b shows 97% accurate classification and 3% misclassification by the proposed model. Therefore, this paper recommends image magnification of 100 × as the best magnification for evaluating the BreakHis dataset. Figure 7c presents the CM for 200 × magnification which recorded 100% prediction for adenosis leading to a total misclassification of 5% and an accurate classification of 95%. However, the proposed model recorded a misclassification of 6% and a correct classification of 94% on the 400 × magnification. The cause of low classification performance recorded by the proposed model on the 400 × magnification was a result of the 81% misclassification of lobular carcinoma that has affected the total performance on the particular magnification as shown in Fig. 7d.

Analysis of ES-CNN and existing study

Table 3 presents a comparative tabulation of the performance of some of the state-of-the-art proposed solutions for classification BHI that used BreakHis, which is the same dataset used to evaluate ES-CNN. In Table 3, ES-CNN exhibits superior performance in histopathological image analysis by integrating patient and magnification dependencies into its architectural design. The image (%) column represents the proportion of total images available at each magnification and how the dataset is distributed across the different magnifications, while the patient (%) column represents proportion of patients contributing to the dataset at each magnification. This ensures that the dataset accounts

Table 3 Comparison of ES-CNN with existing work

Methods	Title	Training/test	Magnification	Image (%)	Patient (%)
Gour et al. [40]	Residual learning-based CNN for breast cancer histopathological image classification	70/30	40 ×	87.40	87.47
			100 ×	87.26	88.15
			200 ×	91.15	92.52
			400 ×	86.27	87.78
Alkassar et al. [41]	Going deeper: magnification-invariant approach for breast cancer classification using histopathological images	70/30	40 ×	99	X
			100 ×	98.5	
			200 ×	98.5	
			400 ×	98	
Li et al. [14]	Classification of breast cancer histopathological images using inter-leaved DenseNet with SENet (IDSNet)	50/20/30	40 ×	89.5	89.1
			100 ×	87.5	85.0
			200 ×	90.0	87.0
			400 ×	84.0	84.5
Sharma and Mehra [42]	Effect of layer-wise fine-tuning in magnification-dependent classification of breast cancer histopathological image	80/20	40 ×	89.31	Not reported
			100 ×	85.75	
			200 ×	83.95	
			400 ×	84.33	
Celik et al. [43]	Automated invasive ductal carcinoma detection based using deep transfer learning with whole-slide images	80/20	Magnification	99.11	89.88
Yari et al. [17]	Deep learning applied for histological diagnosis of breast cancer	6011/1142/406	40 ×	100	Not reported
			100 ×	100	
			200 ×	98.08	
			400 ×	98.99	
			Magnification	99.26	
Liu et al. [15]	Fine-grained breast cancer classification with bilinear convolutional neural networks (BCNNs)	Random 5-folds	40 ×	99.33	Not reported
			100 ×	99.04	
			200 ×	98.84	
			400 ×	98.53	
			Magnification	99.24	
Budak et al. [11]	Computer-aided diagnosis system combining FCN and bi-LSTM model for efficient breast cancer detection from histopathological images	Random 5-folds	40 ×	95.69	Not reported
			100 ×	93.61	
			200 ×	96.32	
			400 ×	94.29	
Mewada et al. [16]	Spectral-spatial features integrated convolution neural network for breast cancer classification	Random 70/30	40 ×	97.58	Not reported
			100 ×	97.44	
			200 ×	97.28	
			400 ×	97.02	
Hao et al. [12]	Breast cancer histopathological images classification based on deep semantic features and gray-level co-occurrence matrix	70/30	40 ×	96.75	96.33
			100 ×	95.21	95.26
			200 ×	96.57	96.09
			400 ×	93.15	92.99
Proposed method	Multi-classification of breast cancer histopathological image using enhanced shallow convolutional neural network	80/20	40 ×	97.41	
			100 ×	96.72	
			200 ×	95.97	
			400 ×	95.97	

Table 4 Computation time taken for processing the proposed ES-CNN

Magnification	Number of images	Total processing time (minutes)
40 ×	5985	42.50
100 ×	6243	138.70
200 ×	6039	43.07
400 ×	5460	42.48

for biological variability, as multiple images from the same patient can inflate the data size but may not add diversity. Considering image (%) approach facilitates the best classification performance across the four magnification levels, achieving multi-classification accuracy of up to 98% while maintaining low computational utilization. Other studies that use only Image% representation includes the work of Mewada et al. [16], Budak et al. [11], Liu et al. [15], Yari et al. [17], and Sharma and Mehra [42]. BreakHis dataset contains histopathological images of breast cancer tissues, captured at four different magnifications 400 ×, 200 ×, 1200 ×, and 40 ×. These magnifications column in Table 3 represent different levels of detail and resolution, enabling the analysis of cancerous tissues at multiple scale. The “random 70/30” in the training/test column means that the dataset was randomized before splitting, while in “70/30” randomization is negligible.

In comparison, while several alternative methodologies demonstrate innovations in feature extraction and transfer learning, they exhibit limitations in both generalization across magnifications and computational efficiency.

First, the ResHist model proposed by Gour et al. [40] achieves commendable accuracies of 84.34% without augmentation and 92.52% with augmentation for binary classification tasks. However, this performance is constrained by its binary classification focus and reliance on data augmentation, which can inflate reported performance metrics without addressing model robustness. Similarly, the work of Alkassar et al. [41] on MSB and MSM approaches achieves high magnification-specific binary classification accuracy, but the model lacks the scalability and versatility seen in ES-CNN for multi-class tasks spanning diverse magnifications.

Second, advanced feature extraction strategies, such as those by Liu et al. [15], Yari et al. [17], and Liu et al. [15], exhibit promise but fall short in leveraging magnification dependencies explicitly. For instance, Yari et al. [17] model achieves up to 98% accuracy for multi-class tasks, which is comparable to ES-CNN. However, its reliance on fine-tuning pre-trained models on ImageNet suggests potential limitations in computational efficiency and contextual specificity compared to the magnification-guided ES-CNN architectural design. Similarly, Liu et al. [15] bilinear CNN method, while effective in binary classification, demonstrates reduced performance for fine-grained multi-classification tasks, indicating its limited adaptability to magnification-dependent nuances.

Finally, approaches leveraging transfer learning and hybrid architectures, such as those by Sharma and Mehra [42], Celik et al. [43], Budak et al. [11], and Mewada et al. [16], present fascinating innovations, particularly in employing pre-trained networks and advanced feature fusion techniques. However, these methods often trade computational efficiency for accuracy or fail to report comprehensive magnification-specific metrics.

For instance, Budak et al. [11] use of Bi-LSTM for sequential data introduces additional complexity, while Mewada et al. [16] integration of spectral and spatial features, although innovative, lacks direct magnification-context adaptability. Thus, ES-CNN stands out as a holistic approach, balancing computational efficiency and classification accuracy across varied magnifications, setting a new benchmark in histopathological image analysis.

Conclusions

This paper proposed a shallowed architecture as an ES-CNN architecture for the multi-classification of BHI with significant improvements in classification accuracy as well as computational efficiency. Having set a trade-off between computational demand and performance, by careful designing of magnification and patient dependencies, the proposed ES-CNN model presents a stable performance. This research accomplishes the following:

1. The ES-CNN model achieved multi-classification accuracy rates of 96%, 95%, 98%, and 96% across image magnifications of $400 \times$, $200 \times$, $100 \times$, and $40 \times$, respectively. These results indicate a high level of accuracy in classifying breast cancer histopathological images compared to existing methods.
2. The proposed ES-CNN model has minimized computational utilization and reduced training time. This has been achieved through a simplified architecture that requires fewer weights and less computation time compared to deep convolutional neural networks (DCNNs).
3. Unlike traditional DCNNs, which often require significant computational resources and extensive training times, the ES-CNN is structured to be more efficient while maintaining high classification accuracy. This efficiency is primarily due to its shallow architecture, which contrasts with deeper models like ResNet or DenseNet that have been used in similar contexts.
4. When compared with other state-of-the-art models using the BreakHis dataset, the ES-CNN shows competitive performance. For instance, some models like those utilizing ResNet or DenseNet architectures have achieved accuracies in the range of 92–99% depending on the specific task and dataset configuration. The performance of ES-CNN has shown to be comparable or superior in terms of accuracy while being more computationally efficient.
5. The manuscript highlights that the ES-CNN model employs a novel approach by integrating shallow convolutional layers that reduce complexity without sacrificing accuracy. This approach contrasts with deeper networks that often rely on transfer learning from large pre-trained models.

The advancement of the proposed method can be considered in the following ways: first, by investigating the robustness and generalizability of the proposed model and by experimenting with other publicly available histopathological images. Second, the proposed model can be improved as lightweight for deployment on smaller handheld devices for user accessibility. Finally, various optimization methods such as weight, layer,

and group normalization can be utilized to improve the convergence time and possibly improve the multi-classification accuracy of the proposed model.

Abbreviations

Adadelta	Adaptive delta
AdaGrad	Adaptive gradient
Adam	Adaptive moment estimation
Adamax	Adam-Max
AI	Artificial intelligence
ANN	Artificial neural network
AUROC	Area under the ROC curve
BHI	Breast histopathological image
BC	Breast cancer
CAD	Computer-aided diagnostic
CL	Convolutional layers
CM	Confusion matrix
CNN	Convolutional neural network
DCNN	Deep convolutional neural network
DL	Deep learning
DNN	Deep neural network
DeCAF	Deep convolutional activation feature
ES-CNN	Enhanced shallow convolutional neural network
ELU	Expected linear unit
FN	False negative
FP	False positive
FCL	Fully connected layer
GAN	Generative adversarial network
GD	Gradient descent
GDX	Gradient descent with momentum and adaptive learning rate
HD-sEMG	High-density surface electromyography
ILSVRC	ImageNet Large Scale Visual Recognition Challenge
LSTM	Long short-term memory
MLP	Multilayer perceptron
MRI	Magnetic resonance imaging/image
MaxPL	Max pooling layer
Nadam	Nesterov-accelerated adaptive moment
PET	Positron emission tomography
PCA	Principal component analysis
RGB	Red, green, and blue
ROC curve	Receiver operating characteristic curve
ROI	Region of interest
ReLU	Rectified linear unit
ResNet	Residual network
RMSprop	Root-mean-square propagation
SGD	Stochastic gradient descent
SCNN	Shallow convolutional neural network
SVM	Support vector machine
TN	True negative
TP	True positive
VA-GAN	Variational autoencoder GAN
VGG	Visual geometry group
WSI	Whole slide image
X	Pixel

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Authors' contributions

MY handled the data collection, architectural design, algorithm design, model implementation, experimentation, and result reporting. The first draft of the manuscript was written by MY. AFDK reviewed the concept, the title, the methods, and the experiments. MAB reviewed the result and discussion and edited the first draft of the manuscript, while MA reviewed the final manuscript. All authors read and approved the final version of the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available via Breast Cancer Histopathological Database (BreakHis) — Laboratório Visão Robótica e Imagem (for.br) [38].

Declarations

Competing interests

The authors declare that they have no competing interests.

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